

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

**Keep in Mind**

**This is a practice test.** It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

**Part A — Vocabulary (8 pts)**

Write the letter of the best description next to each term.

- |                                   |                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1. Determinant <b>D</b>           | (A) The smaller determinant left after crossing out one entry's row and column.                                           |
| 2. Minor <b>A</b>                 | (B) A map $\mathbf{x} \mapsto A\mathbf{x}$ that fixes the origin and sends grid lines to straight, evenly spaced lines.   |
| 3. Cofactor expansion <b>F</b>    | (C) A matrix with $\det = 0$ : its columns are dependent, it collapses space, and it has no inverse.                      |
| 4. Linear transformation <b>B</b> | (D) A number computed from a square matrix; its absolute value is the area (or volume) of the box built from the columns. |
| 5. Orientation <b>G</b>           | (E) The rule $\det(AB) = \det A \det B$ : composing transformations multiplies their scaling factors.                     |
| 6. Singular matrix <b>C</b>       | (F) Computing a determinant along one row: each entry times its signed minor, with the pattern $+$ $-$ $+$ $\dots$        |
| 7. Area/volume factor <b>H</b>    | (G) Whether a transformation keeps or flips the handedness of space — kept when $\det > 0$ , flipped when $\det < 0$ .    |
| 8. Product rule <b>E</b>          | (H) The quantity $ \det A $ , the number every area (or volume) is multiplied by under the transformation $A$ .           |

**Part B — Multiple Choice (12 pts)**

Circle the best answer.

- The determinant of the identity matrix  $I$  is (A) 0 (B) 1 ← (C)  $n$  (D) undefined
- Swapping two rows of a matrix makes  $\det = 0$  (A) leaves  $\det$  unchanged (B) **negates**  $\det$  ← (C) doubles  $\det$  (D)
- Adding a multiple of one row to another row (A) negates  $\det$  (B) scales  $\det$  (C) **leaves**  $\det$  **unchanged** ← (D) makes  $\det = 0$
- A square matrix with  $\det = 0$  is (A) invertible (B) **singular** ← (C) orthogonal (D) symmetric
- The number  $|\det A|$  tells you (A) the trace (B) **how much  $A$  scales area/volume** ← (C) the rank (D) nothing geometric
- The determinant of a product satisfies (A)  $\det(AB) = \det A + \det B$  (B)  $\det(AB) = \det A \det B$  ← (C)  $\det(AB) = \det A - \det B$  (D) no rule

Teacher Note

Item 1:  $\det I = 1$  (unit square/cube has area/volume 1) — the founding rule. Item 2: a swap is a reflection, so it flips sign. Item 3: an elimination step is a shear; same base and height, so  $\det$  is unchanged. Item 4:  $\det = 0 \Leftrightarrow$  dependent columns  $\Leftrightarrow$  no inverse = singular. Item 5:  $|\det A|$  is the area/volume scaling factor. Item 6: the product rule  $\det(AB) = \det A \det B$ .

### Part C — Short Answer & Computation (35 pts)

Show your work.

- Determinant of  $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$ ; area and orientation.  $\det A = (1)(1) - (3)(2) = 1 - 6 = -5$ . **The parallelogram area is  $|\det A| = 5$ . Since  $\det A < 0$ ,  $A$  reverses orientation (it flips the plane).**
- Determinant of  $B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$  by cofactor expansion. **Across the top row (signs  $+$   $-$   $+$ ):  $\det B = 1 \cdot \det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} - 2 \cdot \det \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} + 0 = 1(6 - 0) - 2(0 - 1) + 0 = 6 + 2 = 8$ .**
- Determinant of  $C = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 5 \\ 0 & 2 & 5 \end{bmatrix}$  by elimination, then pivots.  $R_2 - 2R_1$ :  $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 5 \end{bmatrix}$ ; then  $R_3 - 2R_2$ :  $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{bmatrix}$ . **Shears don't change  $\det$ , so  $\det C = \text{product of pivots} = 1 \cdot 1 \cdot 3 = 3$ .**
- $\det A = 6$  ( $3 \times 3$ ): effect of each row operation. (a) **Swap two rows  $\Rightarrow$  sign flips:  $\det = -6$ .** (b) **Scale a row by 3  $\Rightarrow$   $\det$  scales by 3:  $\det = 18$ .** (c) **Add  $2R_1$  to  $R_3$  (a shear)  $\Rightarrow$  unchanged:  $\det = 6$ .** (d)  **$\det(2A) = 2^3 \det A = 8 \cdot 6 = 48$  (every one of the 3 rows is scaled by 2).**
- $\det A = 4$ ,  $\det B = 2$  ( $3 \times 3$ ). (a)  $\det(AB) = \det A \det B = 4 \cdot 2 = 8$ . (b)  $\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{4}$ . (c)  $\det(A^T) = \det A = 4$ . (d)  $\det(A^2) = (\det A)^2 = 16$ .
- Transformation  $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ .  $\det A = (3)(2) - (1)(0) = 6$ . (a) **Unit square  $\rightarrow$  parallelogram of area  $|\det A| = 6$ .** (b) **Every area is multiplied by 6, so the triangle's image has area  $4 \cdot 6 = 24$ .** (c)  **$\det A = 6 > 0$ , so orientation is preserved (no flip).**
- $M = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ . (a)  $\det M = (2)(2) - (1)(4) = 4 - 4 = 0$ . (b) **No —  $\det M = 0$  means  $M$  is singular, so it has no inverse.** (c) **The columns  $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$  and  $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$  are parallel (one is twice the other), so the unit square collapses onto a single line through the origin — a flat region of area 0. That is why the area factor is 0 and the map can't be undone.**

### Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- Elimination step and the determinant. (a) **Adding a multiple of one row to another is exactly a *shear*, and one of the founding rules of determinants is that a shear leaves the determinant unchanged. (It is the same elimination step from Unit 2, which is why we can reduce to triangular form and just read off the pivots without the answer changing.)** (b) **Geometrically, the rows build a box. Adding a multiple of one row to another slides the box sideways parallel to a face: the *base* of the box keeps the same area and the *height* above that base is unchanged, so the volume = base  $\times$  height stays the same. Same volume means the same  $|\det|$ , so  $\det$  does not change.**
- Area factor and  $\det = 0$ . (a) **The transformation sends  $e_1, e_2$  to the columns  $Ae_1, Ae_2$ , and the unit square (spanned by  $e_1, e_2$ ) therefore maps to the parallelogram spanned by those two columns. By the meaning of the determinant, that parallelogram has area  $|\det A|$ . Any region can be built from many tiny squares, and each one is scaled by the same factor  $|\det A|$ , so *every* area is multiplied by  $|\det A|$  — that is why it is called the area factor.** (b) **If  $\det A = 0$ , the image parallelogram has area 0: the two columns are parallel, so the unit square is squashed flat onto a single line. The**

transformation collapses the whole plane onto a line, throwing away a dimension, and there is no way to un-squash a line back into a plane. So  $A$  has no inverse — it is singular.

Teacher Note

**Scoring Part D (5 pts each).** D1: 3 pts (a) the step is a shear  $\Rightarrow$  det unchanged (same elimination step as Unit 2); 2 pts (b) box picture — same base area, same height  $\Rightarrow$  same volume. D2: 3 pts (a) unit square  $\rightarrow$  column parallelogram of area  $|\det A|$ , and tiling makes it the factor for every area; 2 pts (b)  $\det = 0 \Rightarrow$  columns parallel  $\Rightarrow$  square collapses to a line (area 0)  $\Rightarrow$  dimension lost  $\Rightarrow$  not invertible. Accept equivalent wording throughout.